TITLE: Resolving Structural Competition: Real-time Sentence Processing in Heritage Spanish–English Readers

Heritage bilinguals provide a critical test case for understanding how two linguistic systems shape the time course of sentence processing (Polinsky & Scontras, 2020). This study examined how Spanish–English heritage speakers read canonical (SVO) and non-canonical (OVS) transitive sentences with inanimate subject–object pairs, using first-pass duration (FPD) and regression fixation duration (RFD) to distinguish early versus later processing stages. The Competition Model suggests that bilinguals show a variety of patterns in sentence interpretation strategies depending due to the competing cue sets from their languages (Bates & MacWhinney, 1989). Previous research on heritage Spanish-English bilinguals and late bilinguals has demonstrated differences in cue utilization and sentence interpretation compared to monolinguals (Reyes & Hernández, 2006). However, research suggests that even when cues are not available, semantic processing alone can allow for highly accurate sentence interpretation (Mahowald et al., 2023). This is especially the case with inanimate-inanimate (IA-IA) sentences, like in this study, where there is less reversibility of thematic roles without competing with semantic plausibility (Mahowald et al., 2023).

Participants (N=48) completed a within-subjects eye-tracking experiment in which Spanish–English heritage bilinguals read 80 transitive sentences in both English and Spanish in SVO and OVS word orders. Stimuli were fully balanced for number agreement, incorporating all combinations of singular/plural subject–object pairs (sing–sing, sing–pl, pl–pl, pl–sing), with verbs morphologically marked to match the subject. Participants read the IA-IA sentence then were instructed to select the subject between the two NPs (Fig. 1). Language measures such as the LexTALE, LexTALE-ESP, and Bilingual Language Profile were also completed.

We modeled accuracy using a binomial mixed-effects regression and analyzed FPD and RFD with linear mixed-effects models, each including word order, number agreement, and individual difference covariates with random intercepts for participants and items. Across languages, first-pass duration showed no significant effects of Word Order (English: $p = .12$; Spanish: $p = .97$), and no significant early-stage interactions, indicating minimal early detection of syntactic mismatch (Fig. 2). In contrast, regression fixation duration revealed robust later-stage sensitivity to Word Order in English, including significant interactions with sentence position (2ndNP × Word Order: $p < .001$; VP × Word Order: $p = .013$), demonstrating increased rereading for non-canonical structures (Fig. 2). The Spanish RFD model demonstrated different patterns of reading from SVO to OVS by sentence position, especially on the 2ndNP (2ndNP: $p < .001$; 2ndNP × Word Order: $p < .001$; Fig. 3). Additionally, differences in number agreement in OVS sentences are seen on the 2ndNP (2ndNP x Number x Word Order: p = 0.001), leading to lower RFD. This suggests that upon encountering the agreement cue on the VP, participants engage in predictive processing.

Together, these results indicate that heritage bilinguals rely primarily on later-stage conflict resolution, not early structural parsing, when cues compete—consistent with the Unified Competition Model. These findings highlight how bilingual experience shapes the timing and mechanisms of sentence processing across languages.

Figure 1: Presentation of the Stimuli

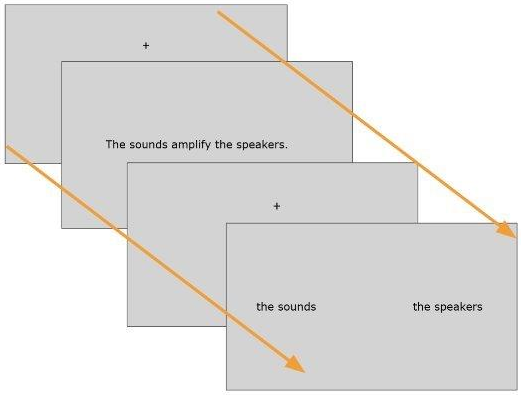


Figure 2: English FPD & RFD Reading Patterns

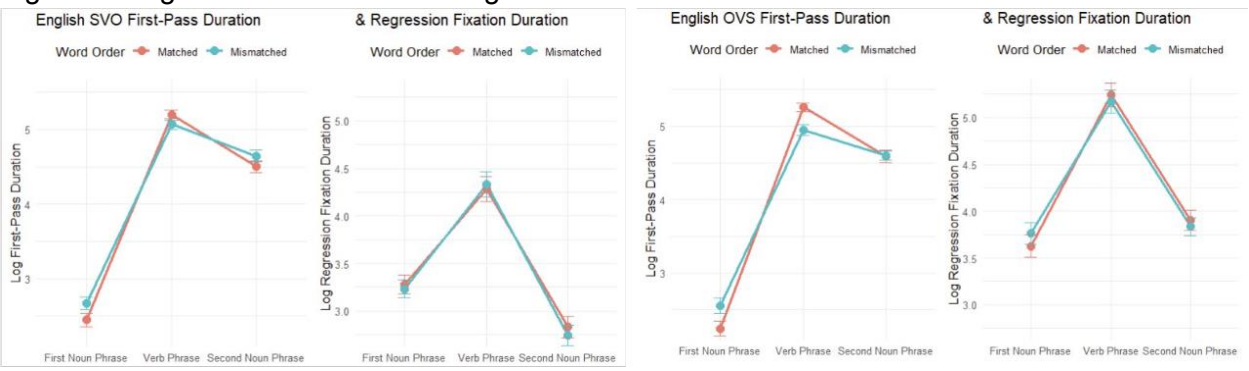
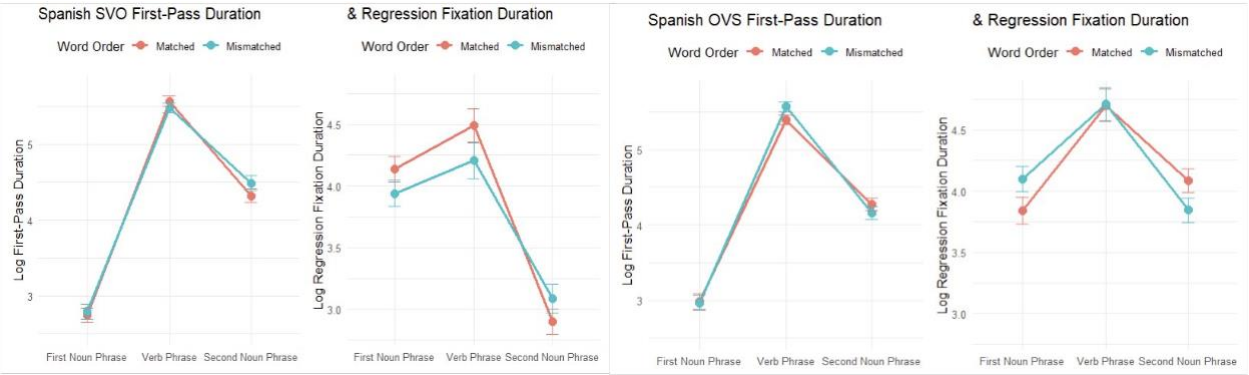


Figure 3: Spanish FPD & RFD Reading Patterns



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